DSBDA Practical Help

i will share you code file of my dsbda practical please explain it as well a give questions for viva with solution

Sure—send the code files (you can upload `.py`, `.ipynb`, screenshots, or paste code here), and I'll help with each practical in this format:

For every program, I'll give:

1. **Aim of the program** (1–2 lines for record book)
2. **Code explanation** (line by line, simple language)
3. **Algorithm / steps**
4. **Output explanation**
5. **Important theory concepts used**
6. **Viva questions with answers** (10–20 per practical)
7. **Possible external examiner questions** ("Why did you use this?", "What is the complexity?", etc.)
8. **Short answer version** (for quick revision before viva)

Since DSBDA practicals usually include things like:

- Data preprocessing
- Handling missing values
- Data visualization
- Correlation / covariance
- Regression
- Classification (Decision Tree / Naive Bayes / KNN)
- Clustering (K-Means)
- Association Rule Mining (Apriori)
- Hadoop / MapReduce basics

...I can tailor the explanation exactly to your syllabus.

Upload the first code file whenever you're ready.

vishnu a1.ipynb
File

I checked your notebook. This is **DSBDA Practical 1 – Data Preprocessing / Understanding Dataset (Iris Dataset)** using Pandas and NumPy.

What your code does

Your notebook performs:
✅ Import libraries
✅ Load CSV dataset
✅ Statistical summary (`describe`)
✅ Check missing values
✅ Display first and last 5 rows
✅ Check duplicates
✅ Dataset info (`info`)
✅ Shape (rows × columns)
✅ Missing values in specific column
✅ Drop a column
✅ Convert categorical labels to numeric values

This is a very common DSBDA viva practical.

---

1) Aim

To perform data preprocessing and exploratory analysis on the Iris dataset using Python libraries.

---

2) Code explanation

Code:

```python
import pandas as pd
import numpy as np
```

Explanation:

- `pandas` → used for handling tables/dataframes
- `numpy` → used for numerical operations

Viva question: Why pandas?
Answer: Pandas is used for data manipulation and analysis.

Code:

```python
df = pd.read_csv('iris.csv')
df.describe()
```

Explanation:

Reads CSV file into dataframe.

`describe()` gives:

- count
- mean
- std deviation
- min
- max
- quartiles

Viva: What is dataframe?
Answer: 2D tabular data structure in Pandas.

Code:

```python
df.isnull()
```

Explanation:

Checks NULL values.

Returns:

- True → missing
- False → present

Code:

```python
print(df.head())
print(df.tail())
```

Explanation:

`head()` → first 5 rows
`tail()` → last 5 rows

Used to inspect dataset.

Code:

```python
df.duplicated()
```

Explanation:

Checks duplicate rows.

True = duplicate row.

Code:

```python
df.info()
```

Explanation:

Shows:

- column names
- datatype
- non-null count
- memory usage

Code:

```python
df.shape
```

Explanation:

Shows:

```
(rows, columns)
```

Example:

```
(150,5)
```

Means:
150 rows
5 columns

Code:

```python
df["sepal.length"].isnull()
```

Explanation:

Checks NULL values in one column only.

Code:

```python
y = df.drop(["petal.length"], axis=1)
print(y)
```

Explanation:

Removes column.

`axis=1` means column operation.

Viva: Why axis=1?
Answer: axis=1 represents columns.

Code:

```python
df['variety'].replace(['Setosa','Virginica'],[0,1],inplace=True)
```

Explanation:

Converts text labels into numbers.

Example:
Setosa → 0
Virginica → 1

This is called **encoding**.

Used in ML algorithms.

Code:

```python
df.isnull().sum()
```

Explanation:

Counts missing values in every column.

Example:

```
sepal.length    0
sepal.width     0
petal.length    2
```

Means petal.length has 2 missing values.

---

3) Algorithm

1. Import libraries
2. Load dataset
3. Display statistics
4. Check null values
5. Show top and bottom rows
6. Detect duplicates
7. Check dataset structure
8. Find dataset shape
9. Remove unwanted columns
10. Convert categorical labels into numeric form

---

4) Viva questions with answers

Basic

Q1 What is data preprocessing?

Cleaning and preparing raw data for analysis.

Q2 Why preprocessing is needed?

To improve data quality and model accuracy.

Q3 What is missing value?

A value not present in dataset (NULL/NaN).

Q4 What is duplicate data?

Repeated rows in dataset.

Q5 What is dataframe?

2D labeled data structure in Pandas.

Q6 Difference between NumPy and Pandas?

NumPy → arrays
Pandas → tables/dataframes

Q7 What does `head()` do?

Shows first 5 rows.

Q8 What does `tail()` do?

Shows last 5 rows.

Q9 What is `describe()`?

Statistical summary.

Q10 What is encoding?

Converting categorical data into numerical form.

Medium

Q11 What is `axis=0`?

Rows

Q12 What is `axis=1`?

Columns

Q13 What is NULL?

Missing value.

Q14 How to remove missing values?

Using:

```python
dropna()
```

Q15 How to fill missing values?

Using:

```python
fillna()
```

Q16 What is Iris dataset?

Famous ML dataset containing flower measurements.

Q17 Features in Iris dataset?

- sepal length
- sepal width
- petal length
- petal width

Target:

- species/variety

Q18 What is inplace=True?

Makes permanent change.

Q19 What does `shape` return?

Rows and columns count.

Q20 What does `info()` show?

Datatype + non-null count + memory.

External examiner may ask:

Why convert categorical into numeric?
→ ML algorithms work better with numerical input.

Difference between cleaning and preprocessing?
→ Cleaning removes errors, preprocessing prepares data for modeling.

What is feature selection?
→ Choosing important columns.

Send your **next practical file**, and I'll explain it in the same viva-ready format.

vishnu a2 (2).ipynb
File

I checked your second notebook. This is **DSBDA Practical 2 – Data Cleaning / Data Wrangling / Outlier Handling** using Pandas and NumPy.

This practical is actually very good for viva because it covers **missing values + invalid data + outliers + normalization**.

Aim

To perform data cleaning and preprocessing on student data by handling missing values, invalid values, outliers, and scaling features.

What your program does

Your code:
✅ Creates student dataset
✅ Assigns grades based on marks
✅ Introduces missing values intentionally
✅ Adds invalid attendance/grade values
✅ Detects errors
✅ Cleans missing values
✅ Corrects invalid values
✅ Detects outliers using IQR
✅ Normalizes attendance values

Code explanation

1) Import libraries

```python
import pandas as pd
import numpy as np
```

Explanation:

- **Pandas** → for table operations
- **NumPy** → numerical calculations/random generation

---

2) Create dataset

```python
np.random.seed(50)
```

Explanation:

Fixes randomness.

Meaning:
Every time same random numbers generated.

Used for reproducibility.

Viva: Why seed is used?
Answer: To generate same random output every run.

Python
```
data = {
    'Student_id': range(1, 51),
```

Creates IDs 1–50.

Python
```
'Name': ['Student_' + str(i) for i in range(1, 51)]
```

Creates:
Student_1
Student_2
Student_3

...

Python
```
'Age': np.random.randint(18,25,size=50)
```

Random age between 18–24.

Python
```
'Gender': np.random.choice(['Male','Female'], size=50)
```

Random gender selection.

Python
```
'Scores': [np.random.randint(50,100,size=3).tolist() for _ in range(50)]
```

Each student gets 3 subject marks.

Example:

```
[88,92,75]
```

Python
```
'Attendance': np.random.randint(20,100,size=50)
```

Attendance percentage.

Python
```
'Grade': np.random.choice(['A','B','C','D','F'],size=50)
```

Random grades.

Python
```python
df = pd.DataFrame(data)
```

Creates dataframe.

---

3) Grade assignment function

Python
```python
def assign_grade(scores):
```

Function for grade.

Average marks calculated:

Python
```python
avg_score=np.mean(scores)
```

If:

90 → A
80 → B
70 → C
60 → D
else F

This is classification.

Python
```python
df['Grade']=df['Scores'].apply(assign_grade)
```

Apply function to every row.

---

4) Add dirty data

Python
```python
df.loc[8,'Age']=np.nan
```

Creates missing value.

NaN = Not a Number / missing.

Python
```python
df.loc[11,'Scores']=None
```

Missing score.

Python
```python
df.loc[9,'Attendance']=105
```

Invalid because attendance >100 impossible.

Python

```python
df.loc[15,'Grade']='Z'
```

Invalid grade.

Allowed:
A,B,C,D,F

Z invalid.

---

5) Detect problems

```python
df.isnull().sum()
```

Counts missing values.

Example:
Age = 3 missing.

```python
invalid_attendance = df[(df['Attendance']<0)|(df['Attendance']>100)]
```

Finds impossible attendance.

```python
invalid_grades = df[~df['Grade'].isin(['A','B','C','D','F'])]
```

Find invalid grades.

~ means NOT.

---

6) Handle missing values

```python
df['Age']=df['Age'].fillna(df['Age'].median())
```

Fill missing age with median.

Why median?
Less affected by outliers.

Viva: Mean vs Median?

Mean → average
Median → middle value

Median better with outliers.

---

7) Fix invalid attendance

```python
lambda x:100 if x>100 else ...
```

If attendance >100 → set 100.

If <0 → set 0.

Otherwise keep same.

---

8) Handle invalid scores

```python
if scores is None:
    return [0,0,0]
```

Missing scores become zero.

```python
return [max(0,min(100,score)) for score in scores]
```

Clamps score between 0 and 100.

Example:

120 → 100
-10 → 0

---

9) Outlier handling using IQR

```python
Q1=df[column].quantile(0.25)
Q3=df[column].quantile(0.75)
```

25th percentile and 75th percentile.

```python
IQR = Q3-Q1
```

Interquartile range.

Formula widget:

$$IQR = Q_3 - Q_1$$

Bounds:

$$\text{Lower} = Q_1 - 1.5(IQR), \quad \text{Upper} = Q_3 + 1.5(IQR)$$

Outside range = outlier.

Then replace with nearest bound.

Viva: Why remove outliers?
Answer: Outliers distort analysis and model performance.

---

10) Scaling / normalization

```python
df['Scaled_Attendance']=
(df['Attendance']-min)/(max-min)
```

Min-Max normalization:

$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$

Range becomes:

0 to 1

Example:
50 → 0.4
80 → 0.8

Viva questions with answers

Basic

1) What is data cleaning?

Removing errors and inconsistencies.

2) What is missing value?

Absent data.

3) What is NaN?

Missing numerical value.

4) What is outlier?

Extreme abnormal value.

5) What is normalization?

Scaling data to fixed range.

Medium

6) Why median for missing values?

Robust against outliers.

7) What is IQR?

Difference between Q3 and Q1.

8) Full form of IQR?

Inter Quartile Range.

9) What is lambda?

Anonymous function.

10) What is apply()?

Applies function row-wise/column-wise.

Advanced

11) Difference between normalization and standardization?

Normalization:
0–1 range

Standardization:
mean=0, std=1

---

12) Why remove invalid values?

Improves quality.

13) Why detect duplicates?

Avoid repeated records.

14) What is feature scaling?

Making features same scale.

15) Why preprocessing important?

Improves ML performance.

This practical is **very viva-important**—expect 5–6 questions from it.

Send **Practical 3** next.

vishnuA3.ipynb
File

I checked your **Practical 3 notebook**. This is **DSBDA Practical 3 – Descriptive Statistics / Grouping / Measures of Central Tendency** using Pandas.

This practical mainly covers:
✅ Grouping data (`groupby`)
✅ Summary statistics (`describe`)
✅ Median
✅ Mode
✅ Percentiles / Quartiles
✅ Analysis on Iris Dataset by species

This is a **theory-heavy viva practical**, so examiners love asking questions from it.

Aim

To perform statistical analysis on datasets using descriptive statistics like mean, median, mode, percentiles, and grouped analysis.

Part 1 – Create dataset

Code:

```python
import pandas as pd

data = {
    'age':[25,30,22,40,55,60,33,28,45,50],
    'income':[50000,60000,45000,70000,80000,90000,65000,62000,75000,85000],
    'age_group':['20-30','30-40','20-30','40-50','50-60','50-60','30-40','20-30','40-50','50-60']
}

df = pd.DataFrame(data)
```

Explanation

Creates dataframe having:

- Age
- Income
- Age group

Example:

age	income	age_group
25	50000	20-30
30	60000	30-40

Part 2 – Groupby + describe

Code:

```python
summary_stats = df.groupby('age_group')['income'].describe()
print(summary_stats)
```

Explanation

Groups data according to age_group.

Then calculates statistics for income.

Example:

20–30 group income stats separately.

30–40 group income stats separately.

`describe()` gives:

- count
- mean
- std
- min
- 25%
- 50%
- 75%
- max

Formula for mean:

$\bar{x} = \frac{\sum x}{n}$

Part 3 – Median

Code:

```python
median_income = df.groupby('age_group')['income'].median()
```

Median = middle value after sorting.

Formula:

$$\text{Median} = \text{middle value of ordered data}$$

Example:
10,20,30,40,50

Median = 30

If even:

10,20,30,40

Median = (20+30)/2 = 25

Viva

Why median preferred sometimes over mean?
Because median is less affected by outliers.

Example:

10,20,30,40,100000

Mean distorted.

Median stable.

Part 4 – Column names

Code:

```python
print(df.columns)
```

Shows:

```
Index(['age','income','age_group'])
```

Useful to inspect dataset columns.

Part 5 – Mode

Code:

```python
mode_age = df['age'].mode()
mode_income = df['income'].mode()
```

Mode = most frequent value.

Formula:

$$\text{Mode} = \text{value with highest frequency}$$

Example:

2,2,2,5,7,7

Mode = 2

Part 6 – Load Iris dataset

Code:

```python
df = pd.read_csv("iris.csv")
print(df.head())
```

Loads Iris dataset.

Columns:

- sepal.length
- sepal.width
- petal.length
- petal.width
- variety

Species:

- Setosa
- Versicolor
- Virginica

Part 7 – Group by species

Code:

```python
summary_stats_species = df.groupby('variety').describe()
```

Calculates statistics species-wise.

Example:

For Setosa:
mean sepal length

For Virginica:
mean sepal length

Separate analysis.

Part 8 – Percentiles

Code:

```python
percentiles = df.groupby('variety').quantile([0.25,0.5,0.75])
```

Meaning

25% → first quartile (Q1)
50% → median (Q2)
75% → third quartile (Q3)

Formula:

$$Q_1 = 25\%,\ Q_2 = 50\%,\ Q_3 = 75\%$$

Used for:

- spread
- outlier detection
- boxplots

Part 9 – Median species wise

Code:

```python
median_values = df.groupby('variety').median()
```

Finds median of all numeric columns per species.

Part 10 – Mode

Code:

```python
mode_width = df['sepal.width'].mode()
```

Most common sepal width.

Important theory for viva

Mean

Average value.

Median

Middle value.

Mode

Most repeated value.

Quartiles

Data split into 4 parts.

Percentiles

Data split into 100 parts.

Standard deviation

Spread of data.

Formula:

$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$

Low SD → close values
High SD → spread values

Viva Questions + Answers

Basic

1) What is descriptive statistics?

Summarizing data numerically.

---

2) Measures of central tendency?

- Mean
- Median
- Mode

---

3) Mean formula?

Average of observations.

---

4) What is median?

Middle value.

---

5) What is mode?

Most frequent value.

---

6) What is percentile?

Value below which percentage observations fall.

---

7) What is quartile?

Divides data into 4 equal parts.

---

8) What is Q1?

25th percentile.

---

9) What is Q2?

50th percentile / median.

---

10) What is Q3?

75th percentile.

Pandas

11) What does `groupby()` do?

Groups similar records.

---

12) What does `describe()` do?

Returns summary statistics.

---

13) What does `mode()` return?

Most frequent value.

---

14) What does `median()` return?

Middle value.

---

15) What does `quantile()` do?

Computes percentile values.

Advanced

16) Difference between mean and median?

Mean = average
Median = middle

---

17) Which is better for skewed data?

Median.

---

18) Why use groupby?

For category-wise analysis.

---

19) What is variance?

Spread around mean.

---

20) What is standard deviation?

Square root of variance.

This practical is mostly **statistics + pandas viva**.

Send Practical 4 and I'll continue in the same format.

Vishnu_A4_Data_Analytics.html
File

I checked your **Practical 4** file Vishnu_A4_Data_Analytics.
This is **DSBDA Practical 4 – Linear Regression (Boston Housing Dataset)** using Pandas, Seaborn, Matplotlib and Scikit-learn.

This is a **very important ML practical**—expect viva on:
✅ Regression
✅ Train/Test split
✅ Correlation matrix
✅ MSE
✅ R^2 score
✅ Model fitting & prediction

Aim

To build a Linear Regression model for house price prediction and evaluate its performance using statistical metrics.

What your code does

Your program:

✅ Imports ML libraries
✅ Loads Boston Housing dataset
✅ Displays dataset info
✅ Checks missing values
✅ Draws correlation heatmap
✅ Splits dataset into training/testing
✅ Trains Linear Regression model
✅ Predicts prices
✅ Calculates MSE and R^2 score
✅ Visualizes Actual vs Predicted values

Code Explanation

1) Import libraries

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Explanation:

- Pandas → dataset handling

- NumPy → numerical operations
- Seaborn → heatmap
- Matplotlib → graph plotting
- train_test_split → split data
- LinearRegression → ML model
- metrics → evaluate model

---

2) Load dataset

```python
df = pd.read_csv("Assignment-A4-BostonHousing.csv")
df.head()
```

Loads CSV file.

Shows first 5 rows.

Dataset columns include:

- crim → crime rate
- zn → residential land
- rm → rooms
- age → age of house
- tax → tax rate
- lstat → lower status population
- medv → median house price (target)

Viva: What is target variable?
Answer: Variable we want to predict (`medv`).

---

3) Dataset info

```python
print(df.columns)
print(df.info())
print(df.describe())
```

columns

Shows feature names.

info()

Shows:

- datatype
- non-null count
- memory usage

describe()

Shows:

- mean
- std
- min
- max
- quartiles

---

4) Missing values

```python
df.isnull().sum()
```

Checks null values.

Your dataset:

All columns = 0 null values.

Meaning clean dataset.

---

5) Correlation matrix

```python
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

Creates correlation heatmap.

Shows relationship between variables.

Correlation formula:

$r = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$

Range:

- +1 → strong positive
- 0 → no relation
- −1 → strong negative

Example:

Rooms ↑ → price ↑

Positive correlation.

Viva: Why correlation matrix?
Answer: To identify feature relationships.

---

6) Split data

```python
X = df.drop('medv', axis=1)
y = df['medv']
```

X = input features

y = output target

```python
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.2,random_state=42)
```

Split:

80% training
20% testing

Viva: Why split data?
Answer: To test model on unseen data.

---

7) Create model

```python
lr = LinearRegression()
```

Creates regression model.

Linear equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where:

- y = prediction
- β = coefficients

---

8) Train model

```python
lr.fit(X_train,y_train)
```

Model learns relationship.

This is called **training/fitting**.

9) Predict

```python
y_pred = lr.predict(X_test)
```

Predicts house prices.

---

10) Evaluate model

```python
mean_squared_error(y_test,y_pred)
```

MSE formula:

$$MSE = \frac{1}{n}\sum(y-\hat{y})^2$$

Lower MSE = better.

Your output:

```
24.29
```

Good.

```python
r2_score(y_test,y_pred)
```

R^2 formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Your output:

```
0.668
```

Meaning:

66.8% variance explained by model

Pretty decent.

---

11) Scatter plot

```python
plt.scatter(y_test,y_pred)
```

Plots actual vs predicted.

If model perfect:

Points lie on straight line.

Red line = ideal prediction line.

Viva Questions + Answers

Basic

1) What is regression?

Predicting continuous values.

Example:
price prediction.

---

2) Difference between classification and regression?

Classification → categories
Regression → continuous number

---

3) What is dependent variable?

Output variable (`medv`)

---

4) Independent variable?

Input features.

---

5) What is Linear Regression?

ML algorithm fitting straight-line relationship.

Medium

6) Why train_test_split?

To evaluate performance.

---

7) What is model fitting?

Training model on data.

---

8) What is prediction?

Estimating output.

---

9) What is correlation?

Relationship between variables.

---

10) Range of correlation?

-1 to +1

Advanced

11) What is MSE?

Average squared error.

---

12) Lower MSE better or higher?

Lower.

---

13) What is R^2 score?

Explained variance.

---

14) Best R^2 value?

1

---

15) What is overfitting?

Model memorizes training data.

---

16) What is underfitting?

Model too simple.

---

17) What assumptions Linear Regression makes?

- linearity
- independence
- normality
- constant variance

---

18) Why scatter plot?

To compare actual vs predicted.

---

19) What does fit() do?

Trains model.

---

20) What does predict() do?

Generates prediction.

This practical is **one of the most viva-important** in DSBDA.

Send Practical 5 next.

Sources

Vishnu_A5.html
File

I'm ready for **Practical 5**, but it looks like the file/code didn't come through on my side.

Please either:

- **Upload the notebook/file** (`.ipynb`, `.py`, screenshot), **or**
- **Paste the code here**

Once you send it, I'll explain it in the same format:
✅ Aim
✅ Code explanation line-by-line
✅ Theory concepts
✅ Algorithm
✅ Output explanation
✅ **15–20 Viva questions with answers**
✅ Quick revision notes for exam

Send **Practical 5** whenever you're ready.